

Выполним эквивалентное преобразование схемы и определим комплексные нагрузки цепи

$$\omega = 2\pi f = 2\pi \cdot 50 = 314 \text{ рад/с}$$

$$E_1 = \frac{U_m}{\sqrt{2}} e^{j\varphi_k} = 17.678 e^{j\frac{\pi}{4}} \text{ В}$$

$$L_1 = \frac{L}{\sqrt{2}} e^{j\varphi} = 10.602 e^{j\frac{\pi}{4}} \text{ Г}$$

$$E_2 = \frac{U}{Z_{\text{вн}}} = \frac{102.45 + j27.45}{0.1} = 1024.5 + j274.5 \text{ В}, R_{\text{вн}} = \frac{1}{0.1} = 10 \Omega$$

$$E_{\text{eq}} = E_1 - E_2 = 16.02 + j7.42 - (1024.5 + j274.5) = -86.43 - j29.98 \text{ В}$$

$$Z_{\text{вн}} = (R_{\text{вн}} + jX_{\text{вн}}) + jX_{L1} = (5 + j0) + j2 = 5 + j2 \Omega = 5.387 \angle 59^\circ \text{ deg}$$

Рассчитаем эквивалентные сопротивления ветвей цепи

$$X_L = \omega L_1 = 314.1593 \cdot 0.006 = 1.885 \Omega, X_C = \frac{1}{\omega C} = \frac{1}{314.159 \cdot 0.00004} = 67.726 \Omega$$

$$X_L = \omega L_2 = 314.1593 \cdot 0.11 = 34.558 \Omega, X_C = \frac{1}{\omega C} = \frac{1}{314.159 \cdot 0.000005} = 90.946 \Omega$$

$$X_C = \omega L_3 = 314.1593 \cdot 0.02 = 21.991 \Omega, X_C = \frac{1}{\omega C} = \frac{1}{314.159 \cdot 0} = 0 \Omega$$

$$Z_1 = R_1 + jX_L + jX_C = 4 + j1.885 + j67.726 = 4 + j69.611$$

$$Z_2 = R_2 + jX_L + jX_C = 16 + j34.558 + j90.946 = 16 + j125.504$$

$$Z_3 = R_3 + jX_L + jX_C = 17 + j21.991 + j0 = 17 + j21.991$$

Находим эквивалентное сопротивление нагрузки и входное сопротивление цепи

$$Z_{\text{вн}} = \frac{Z_1 Z_2}{Z_1 + Z_2} = \frac{(4 + j69.611)(16 + j125.504)}{20 + j127.495} = 31.15 + j14.08$$

$$Z_{\text{eq, load}} = Z + Z_{\text{вн}} = 4 + j69.611 + (31.15 + j14.08) = 35.15 + j83.691$$

$$Z_{\text{in}} = Z_{\text{eq, in}} + Z_{\text{eq, load}} = 15 + j2 + (35.15 + j83.691) = 50.15 + j85.691$$

Определим значения токов

$$I_1 = \frac{E_1}{Z_{\text{in}}} = \frac{17.678 e^{j\frac{\pi}{4}}}{50.15 + j85.691} = 0.67 - j1.06 \text{ А}$$

$$I_2 = I_1 \frac{Z_1}{Z_1 + Z_2} = 0.67 - j1.06 \cdot \frac{4 + j69.611}{20 + j127.495} = 0.67 - j0.29 \text{ А}$$

$$I_3 = I_1 - I_2 = 0.67 - j1.06 - (0.67 - j0.29) = -0.34 - j0.77 \text{ А}$$

Выполняем проверку расчетов по закону Кирхгофа

$$\Delta U = 0 + j0, \Delta U_1 = 0 + j0, \Delta U_2 = -0 + j0$$

$$\Delta P = 0\%, \Delta Q = 0\%$$